Synthesis of mesoporous SiO2-MoS2 core-shell particles

"How can mesoporous SiO2-MoS2 core-shell particles be synthesized for TENGs?" *Generated by LeapSpace*

Mesoporous **SiO_2–MoS_2 core-shell particles** for TENGs are most plausibly made by a **two-step core formation + shell growth + template removal** route: first prepare monodisperse **SiO_2 cores**, then grow a **mesoporous or nanosheet MoS_2 shell** on the surface by hydrothermal/solvothermal assembly, while using a surfactant or pore-directing agent to control shell mesoporosity and thickness. The retrieved evidence does not show a single TENG-specific synthesis recipe for this exact composition, but it does provide clear, transferable routes from closely related mesoporous silica core-shell systems and MoS_2 shell/core nanostructures. [64] [65] [66] [67] [68]

Practical synthesis route

Step	What to do	Why it matters for mesoporous SiO_2–MoS_2 particles
1. Make SiO_2 cores	Prepare spherical **SiO_2 cores** by a Stöber-type sol-gel route or related silica-core method [65] [66] [69]	Gives a uniform, mechanically stable core that can be coated consistently [70] [71]
2. Add a structure-directing agent	Use a surfactant/template such as **CTAB, P123, or anionic surfactant systems** to generate mesoporosity in the shell [64] [65] [66] [69] [70] [72]	Creates ordered or disordered mesopores in the outer layer instead of a dense coating [65] [70]
3. Grow the MoS_2 shell	Form the outer MoS_2 layer by a **hydrothermal reaction** that promotes nanosheet self-assembly on the particle surface [67] [68] [73]	Produces a conformal MoS_2 shell rather than free MoS_2 particles [68] [73]
4. Tune shell structure	Adjust precursor concentration, reaction time, and temperature to control **shell thickness, coverage, and pore size** [70] [74] [75] [76]	Lets you balance surface area against structural continuity [70] [74]
5. Remove template if needed	Calcine or wash out the organic template to open mesopores [65] [66] [69]	Leaves an accessible mesoporous shell, which is useful for TENG surface area and charge trapping [70] [77]

What the evidence implies for the SiO_2–MoS_2 case

- **SiO_2 cores** are usually formed first because silica is easy to make as monodisperse spheres and can act as a robust template for later shell growth. [65] [66] [69]

- **Mesoporosity** is generally introduced by surfactant-templated silica chemistry; in core-shell silica systems, pore size and pore distribution are controlled by the surfactant type, pore-swelling agents, temperature, and TEOS feeding. [64] [65] [74] [78]
- **MoS_2 shell formation** is most directly supported by hydrothermal growth of MoS_2 nanosheets on particle surfaces, which yields a rough, high-area outer shell. [67] [68]
- For a TENG-oriented design, the most useful target morphology is a **uniform SiO_2 core with a thin, continuous, mesoporous or nanosheet-based MoS_2 shell**, because that combination is the one repeatedly associated with stable interfacial structure and enhanced functional performance in related core-shell systems. [68] [70] [73] [77]

Best-supported design knobs

- **Core size:** adjust the silica core diameter to control total particle size and contact behavior. [70] [71]
- **Shell thickness:** tune MoS_2 precursor loading and growth time to avoid patchy coating or an overly thick shell. [70] [76]
- **Pore size / pore density:** use surfactant template choice and swelling agents to shift the mesopore structure. [65] [74]
- **Shell continuity:** prioritize a conformal shell, since the core-shell functional benefits depend on full surface coverage. [68] [73]

Bottom line: the most defensible synthesis strategy is **templated SiO_2 core preparation followed by hydrothermal MoS_2 shell growth under mesostructure-directing conditions**, with the shell porosity controlled by surfactant/template chemistry and growth conditions rather than by a separate post-treatment alone. [64] [65] [66] [67] [68]

Confidence level: **Medium** — the retrieved evidence strongly supports the building blocks of the synthesis route, but it does not provide a single direct, TENG-specific mesoporous SiO_2–MoS_2 protocol.

References

64. Ishii, H., Ikuno, T., Shimojima, A., Okubo, T. (2015) Preparation of core-shell mesoporous silica nanoparticles with bimodal pore structures by regrowth method. *Journal of Colloid and Interface Science*. https://doi.org/10.1016/j.jcis.2015.01.057

65. Kim, J.H., Yoon, S.B., Kim, J.-Y., (...), Yu, J.-S. (2008) Synthesis of monodisperse silica spheres with solid core and mesoporous shell: Morphological control of mesopores. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*. https://doi.org/10.1016/j.colsurfa.2007.04.145

66. El-Toni, A., Khan, A., Ibrahim, M., (...), Sato, T. (2012) Synthesis of double mesoporous core-shell silica spheres with tunable core porosity and their drug release and cancer cell apoptosis properties. *Journal of Colloid and Interface Science*. https://doi.org/10.1016/j.jcis.2012.04.006

67. Guo, K., Shao, X., Chen, M., (...), Wang, L. (2026) Temperature-dependent tribological mechanisms of MoS2@SiO2 core-shell lubricants from 200 to 800 °C: Experiments and molecular dynamics. *Results in Engineering*. https://doi.org/10.1016/j.rineng.2026.112693

68. Jiang, X., Niu, H., Li, J., (...), Fan, B. (2023) Construction of core-shell structured SiO2@MoS2 nanospheres for broadband electromagnetic wave absorption. *Applied Surface Science*. https://doi.org/10.1016/j.apsusc.2023.157355

69. Tian, H., Chen, J., He, J., Liu, F. (2015) Pd-loaded magnetic mesoporous nanocomposites: A magnetically recoverable catalyst with effective enrichment and high activity for DDT and DDE removal under mild conditions. *Journal of Colloid and Interface Science*. https://doi.org/10.1016/j.jcis.2015.07.024

70. Baliś, A., Zapotoczny, S. (2018) Tailored synthesis of core-shell mesoporous silica particles—optimization of dye sorption properties. *Nanomaterials*. https://doi.org/10.3390/nano8040230

71. Eurov, D.A., Kurdyukov, D.A., Kirilenko, D.A., (...), Golubev, V.G. (2015) Core–shell monodisperse spherical mSiO2/Gd2O3:Eu3+@mSiO2 particles as potential multifunctional theranostic agents. *Journal of Nanoparticle Research*. https://doi.org/10.1007/s11051-015-2891-y

72. Mesa, M., Guth, J.-L., Sierra, L. (2005) Micron-sized spherical core-shell particles of mesoporous silica suitable for HPLC applications. *Studies in Surface Science and Catalysis*. https://doi.org/10.1016/s0167-2991(05)80574-9

73. Liao, J., Qiu, J., Wang, G., (...), Wang, W. (2021) 3D core-shell Fe3O4@SiO2@MoS2 composites with enhanced microwave absorption performance. *Journal of Colloid and Interface Science*. https://doi.org/10.1016/j.jcis.2021.07.032

74. Zhang, J., Li, X., Rosenholm, J.M., Gu, H.-C. (2011) Synthesis and characterization of pore size-tunable magnetic mesoporous silica nanoparticles. *Journal of Colloid and Interface Science*. https://doi.org/10.1016/j.jcis.2011.05.038

75. Chen, Q., Ge, Y., Granbohm, H., Hannula, S.-P. (2018) Effect of ethanol on Ag@Mesoporous silica formation by in situ modified stöber method. *Nanomaterials*. https://doi.org/10.3390/nano8060362

76. Arman, M.M., Ramadan, R. (2023) Spherical SiO2 growth on LaFeO3 perovskite to create core–shell structures for Cd(II) adsorption on its surface. *Journal of Materials Science: Materials in Electronics*. https://doi.org/10.1007/s10854-023-10721-2

77. Zhang, S., Du, M., Kuang, J., (...), Zhou, W. (2019) Surface-defect-rich mesoporous NH2-MIL-125 (Ti)@Bi2MoO6 core-shell heterojunction with improved charge separation and enhanced visible-light-driven photocatalytic performance. *Journal of Colloid and Interface Science*. https://doi.org/10.1016/j.jcis.2019.07.021

78. Baliś, Andrzej, Zapotoczny, Szczepan (2018) Tailored Synthesis of Core-Shell Mesoporous Silica Particles—Optimization of Dye Sorption Properties. *Nanomaterials*. https://doi.org/10.3390/nano8040230